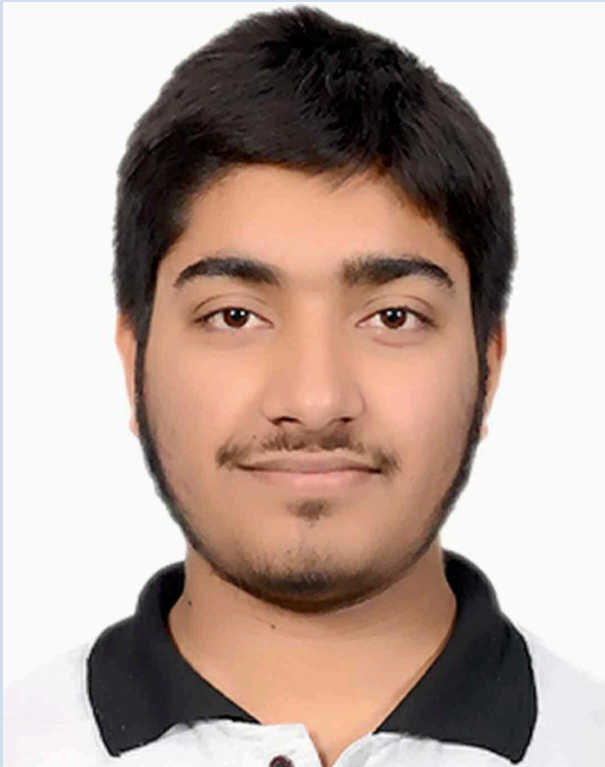




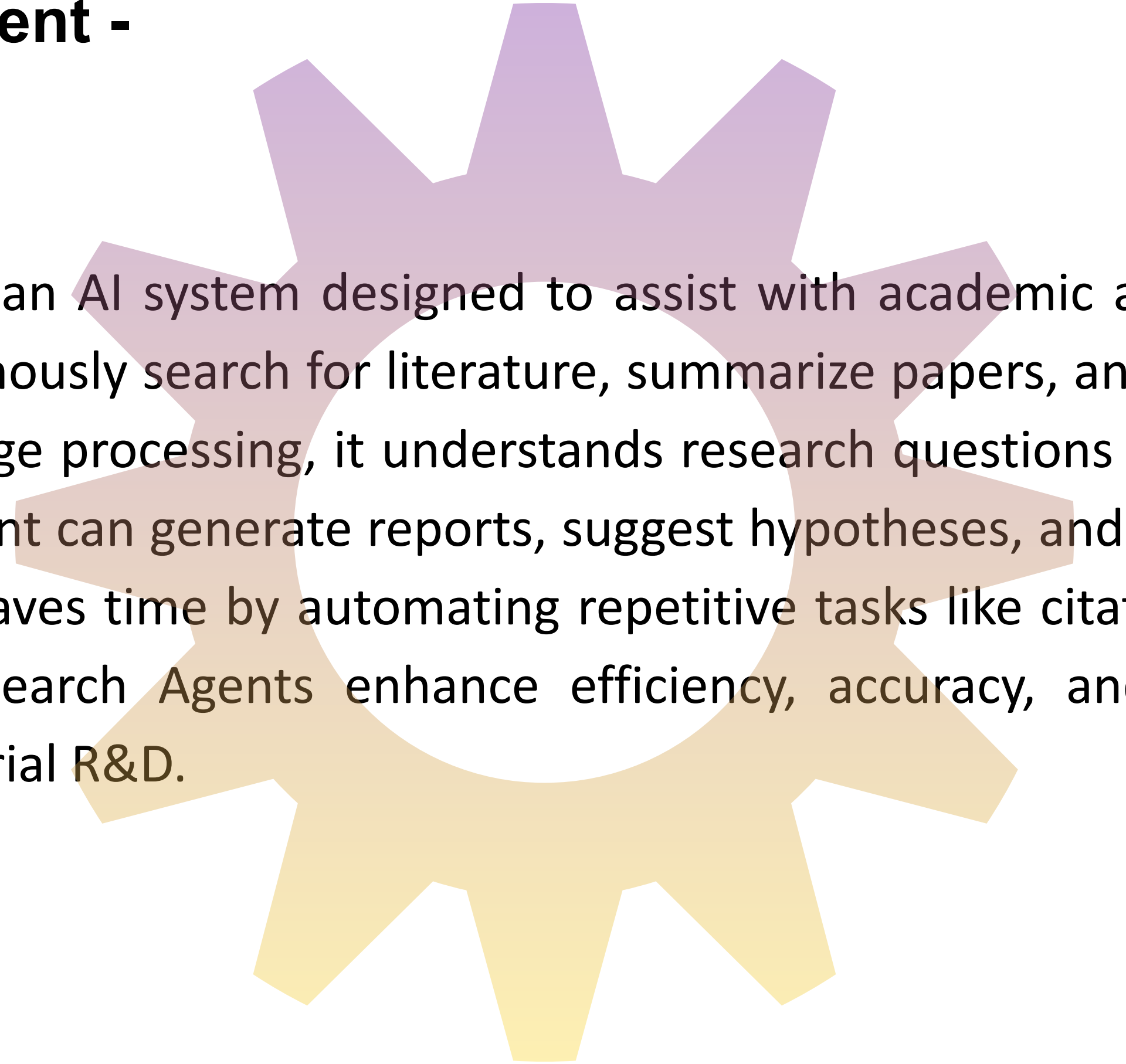
IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Domain of Project – Education

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Problem Statement -



A Research Agent is an AI system designed to assist with academic and scientific research tasks. It can autonomously search for literature, summarize papers, and organize references. Using natural language processing, it understands research questions and retrieves relevant information. The agent can generate reports, suggest hypotheses, and even draft sections of research papers. It saves time by automating repetitive tasks like citation management and data extraction. Research Agents enhance efficiency, accuracy, and innovation in both academic and industrial R&D.

AI Frameworks and Tooling Overview

Core AI Technologies

IBM watsonx.ai foundation models for enterprise-grade AI
Meta Llama 3 70B Instruct model for instruction tuning and reasoning
Agentic, multi-agent architecture to orchestrate specialized AI agents

Development Frameworks

LangChain for prompt management, tool orchestration, and chains
LangGraph for stateful, graph-based agent workflows
Python 3.12 as primary language for implementation and scripting

Supporting Tools

Jupyter Notebook for experimentation, prototyping, and demos
Python Dotenv for secure environment variable management
Logging framework to improve observability and debugging
UUID-based memory tracking for robust session-level traceability

Search, Retrieval, and Memory



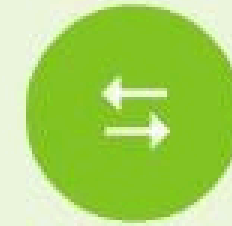
Search and Retrieval

Tavily Search API for structured, targeted web queries
Real-time retrieval of up-to-date web information
Supports evidence gathering for research-grade responses



Memory and State Management

MemorySaver Checkpointer for durable, persistent agent state
InMemoryStore for low-latency, in-session data storage
LangGraph-based state management to maintain agent context flow



Reliability and Traceability

Session-scoped memory tracking using UUID-based identifiers
Structured logging of agent steps, intermediate data, and outputs
Improved debuggability and auditability across research sessions

Research Pain Points

Visualizes key workflow bottlenecks slowing research cycles and raising cognitive load, guiding why an agentic AI assistant is needed.

Discovery

Researchers spend extensive time searching, scanning, and filtering vast literature, manually checking relevance and credibility before they can even begin deeper conceptual or visual exploration.

Drafting

Authoring involves repetitive drafting and redrafting of narrative and visual reports, consuming effort that could be redirected to framing insights, structuring arguments, and refining design-centric communication.

Review

Peer review and quality checks are labor-intensive, requiring meticulous validation of correctness, clarity, and visual consistency across documents, slides, and supplementary design assets.

Control

Version control and revision management introduce heavy effort, as teams track changes, reconcile feedback, and maintain consistent file states across documents, visuals, and collaborative workspaces.

Agentic Research Workflow Overview

Step1

End-to-End Interaction Flow

User submits a query or research topic as the entry point
Planning Agent designs a research outline and strategy from the request
Research Agent calls the Tavily Search API to collect supporting evidence
Writer Agent composes an initial draft report using the plan and gathered results

Step2

Editorial Decisions and Outcomes

Editor Agent assesses the draft and selects an appropriate decision path
Accept: validated final report is returned to the user as output
Reject: workflow stops and provides a clear failure reason
Review: initiates a review and critique loop for iterative refinement

Review Loop and Reinforcement Flow

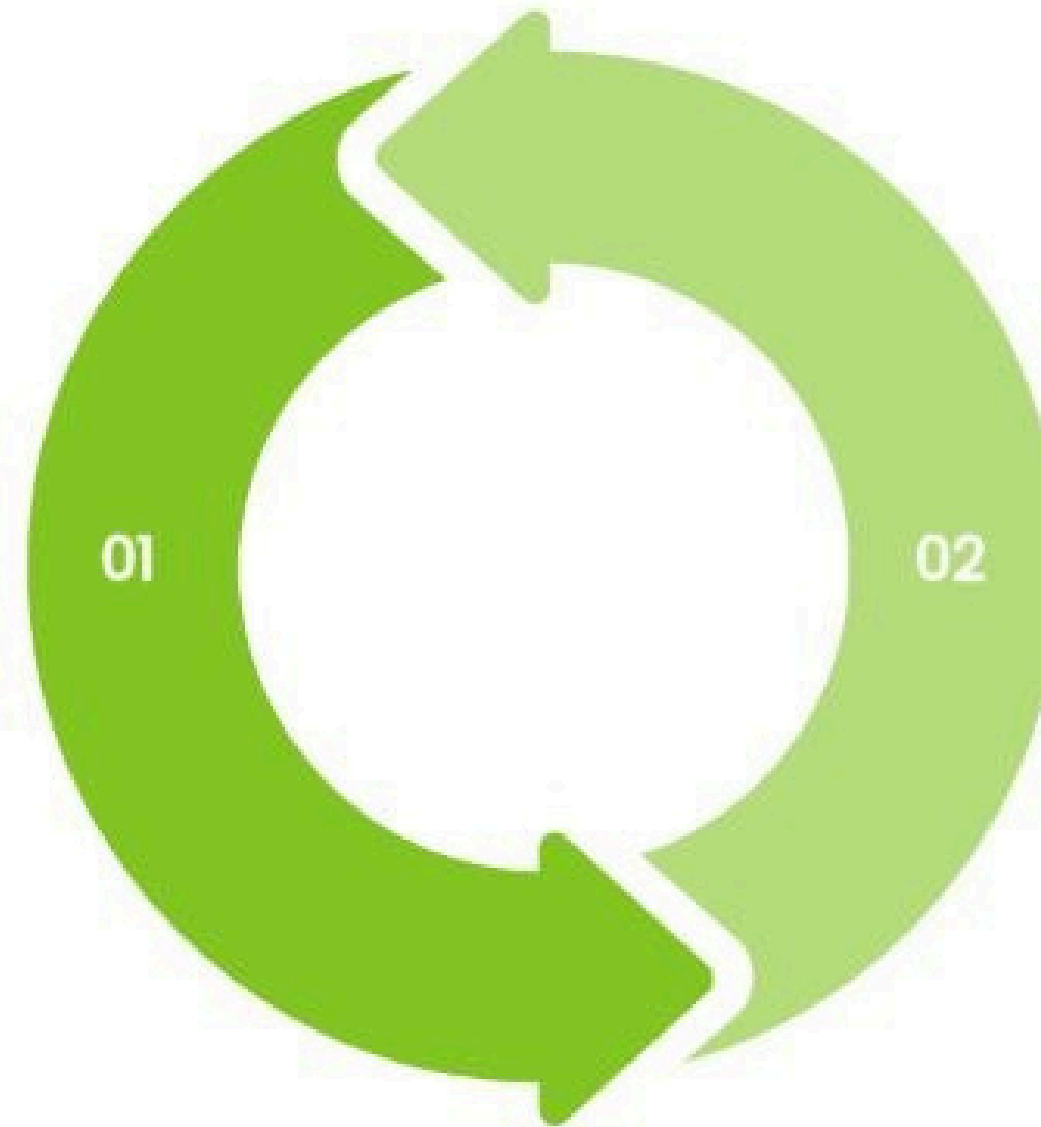
Multi-Agent Review and Critique

Editor triggers the Review flow if draft quality is inadequate

Reviewer Agent evaluates coherence, completeness, and document structure

Critic Agent identifies weak reasoning, missing arguments, and insufficient evidence

Each agent produces focused feedback from its specific evaluation perspective



Iterative Rewrite and Acceptance Loop

Feedback is synthesized into structured revision instructions

Writer Agent regenerates an improved draft guided by these instructions

Loop repeats until the Editor explicitly accepts or rejects the draft

Final output is a validated research report with a documented multi-stage review history

Benefits of Agentic AI in This System



Core benefits for research workflows

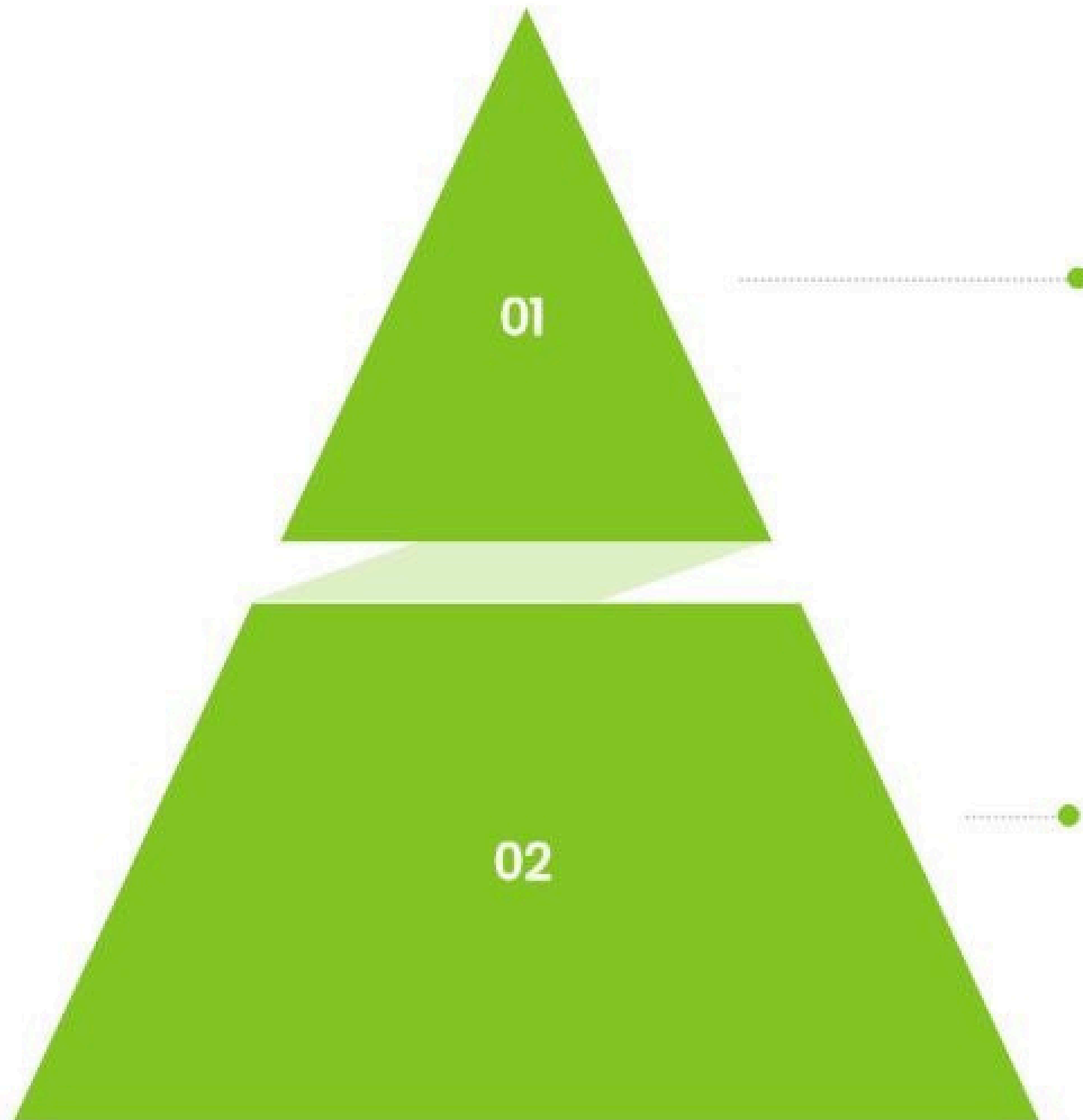
- Executes complex workflows autonomously with minimal human input
- Continuously improves via built-in review, critique, and revision loops
- Reduces manual effort in research, drafting, editing, and validation tasks
- Elevates report quality through multi-stage refinement pipelines
- Produces more reliable results compared to single-pass AI generations



System-level impact on the solution

- Emulates human research teams through specialized, coordinated agent roles
- Enables scalable, repeatable research processes across multiple projects
- Strengthens trust and adoption of AI-generated research deliverables

Agent Components in the Stack



Functional Agent Roles

Planner Agent: defines research goals, strategy, and structured outline
Research Agent: runs Tavily-powered search, collects and aggregates evidence
Writer Agent: turns collected insights into structured research report drafts
Reviewer Agent: assesses draft quality, coverage completeness, and clarity of exposition
Critic Agent: identifies logical gaps, weak arguments, and methodological issues
Editor Agent: makes final accept / reject / revise decisions on the research output

Architectural Responsibilities

Each agent encapsulates a narrow, well-defined research responsibility
Agents interact through LangGraph-managed state transitions for orchestration
Pipeline is modular, allowing agents to be added, removed, or reconfigured independently
Design supports extensible, composable research workflows across the full stack

Tool Gaps

Current setups

Single-step workflows

Most tools run as single agents with one-off LLM calls, producing static outputs that lack iteration, refinement, and structured internal validation cycles.

Weak content validation

Generated reports are rarely cross-checked against sources, leaving factual claims, citations, and reasoning chains largely unverified and error-prone.

No improvement loop

Outputs are created once without autonomous review, so tools cannot detect weaknesses, propose revisions, or iteratively enhance clarity and rigor.

VS

Practical impact

Error-prone outcomes

Lack of verification leads to higher hallucination risk, logical inconsistencies, and subtle inaccuracies that can mislead research and downstream design work.

Missing QA systems

There is no systematic quality assurance layer, such as automated critique, multi-agent review, or scoring to enforce reliability and methodological soundness.

Manual user burden

Researchers and designers must manually inspect, fact-check, and rewrite AI drafts, consuming time, increasing cognitive load, and reducing productivity gains.

Short-Term Enhancements



● Ingestion

Implement robust PDF research paper ingestion and parsing to accurately capture structure, figures, and references, enabling rapid analysis tailored to creative research workflows and visual documentation needs.

● Summarization

Deliver concise research paper summarization and key-point extraction that highlights methods, findings, and visual implications, helping creatives quickly evaluate relevance before deeper reading or design exploration.

● Citations

Introduce automatic citation generation for all referenced sources, streamlining academic documentation in design reports, case studies, and project decks, while reducing repetitive manual citation work.

● Formatting

Provide precise APA and IEEE reference formatting support to align with academic standards, tightening integration with research workflows for students, design teams, and creative professionals.

GitHub Link



Medium-Term and Advanced Capabilities



Medium-term enhancements

RAG-based knowledge base to improve domain-specific accuracy

Semantic search for concept-level information retrieval

Vector database integration for fast and scalable similarity search

Automated detection of research trends across large corpora

Topic clustering to group related studies and insights

Advanced future scope

Multi-modal research agent handling text, images, graphs, and tables

Support for generating and refining research hypotheses

Automatic creation of structured literature reviews

Assistance in drafting research papers, including sections and citations





Thank You!

Thank you for your time and interest.